

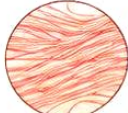


MATRIX – THE DEFINING FEATURE OF CONNECTIVE TISSUE

CLASS 9
SCIENCE

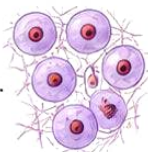
- Matrix is the **non-living intercellular substance** present between the cells of connective tissue.
- In connective tissues, **cells are loosely packed** and embedded in this matrix.

THE MATRIX CAN BE DIFFERENT IN DIFFERENT CONNECTIVE TISSUES

TYPE OF MATRIX	CONNECTIVE TISSUE	FUNCTION	EXAMPLE / LOCATION
 Fluid (Liquid)	 Blood	Transport of materials (gases, nutrients, wastes, hormones) throughout the body.	 Inside blood vessels and heart
 Jelly-like (Flexible)	 Cartilage	Provides flexibility , cushioning and acts as a shock absorber.	 Tip of nose, outer ear, between vertebrae, ends of long bones (joints)
 Hard and Rigid	 Bone	Provides strength , support and protection to the body.	 Skull, ribs, vertebrae, limb bones
 Fibrous	 Tendon & Ligament	Provides great strength and connects one structure to another.	 Tendons connect muscle to bone; ligaments connect bone to bone

WHY IS THE MATRIX IMPORTANT?

- ✓ Holds the cells together.
- ✓ Determines the strength, flexibility and function of the tissue.
- ✓ Gives each connective tissue its special properties.



NCERT KEY CONCEPT

“ Different connective tissues have different types of matrix, and the nature of the matrix decides the function of the tissue. ”



Remember: Same type of cells + **Different type of matrix** = **Different connective tissue** with different function.



FLUID CONNECTIVE TISSUE – BLOOD

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Blood is the only **fluid connective tissue** in our body.
Its matrix is a fluid called **plasma**.

COMPONENTS OF BLOOD

Component	Description	Function
Plasma (≈55% of blood)	Pale yellow, watery fluid. Contains water, proteins, salts, hormones, enzymes, dissolved nutrients and wastes.	Acts as a transport medium for cells and other substances.
RBCs (Erythrocytes) (≈44% of blood)	Red-coloured cells; contain haemoglobin (iron-rich protein); no nucleus; biconcave disc shape; life span ≈ 120 days.	Carry oxygen from lungs to body cells and some carbon dioxide from cells to lungs.
WBCs (Leukocytes) (<1% of blood)	Larger than RBCs; colourless; have nucleus; different types.	Larger than RBCs; against infections; engulf and destroy harmful microbes.
Platelets (Thrombocytes) (<1% of blood)	Very small cell fragments (not complete cells); no nucleus.	Help in blood clotting at the site of injury to stop bleeding.

EVERYDAY OBSERVATIONS & EXPLANATIONS

Observation	Explanation
A small cut on skin bleeds and then clots.	Platelets release chemicals that convert fibrinogen to fibrin; blood clots and seals the wound.
Infection area becomes red, swollen and sometimes has pus.	WBCs rush to the site to fight microbes; inflammation causes redness and swelling; dead WBCs and microbes form pus.
Face turns red and we breathe faster during exercise.	Muscles need more oxygen; RBCs deliver more oxygen and CO ₂ removal increases; breathing rate rises.
Blood is always red in colour.	Due to haemoglobin in RBCs that gives blood its red colour.

FUNCTIONS OF BLOOD

- Transport of oxygen, carbon dioxide, nutrients, hormones, wastes and heat.
- Defence against infections.
- Helps in blood clotting to prevent excessive blood loss.
- Maintains body temperature and pH balance.



STRUCTURE OF BLOOD



Plasma
(≈55%)

Buffy coat
(WBCs + Platelets)
(<1%)

RBCs
(≈44%)



Plasma – Pale yellow, watery fluid; contains water, proteins, salts, hormones, nutrients, wastes.



WBCs – Defend the body from infections.
Platelets – Help in blood clotting.



RBCs – Contain haemoglobin; transport oxygen.

KEY POINTS

- Blood flows in blood vessels throughout the body.
- Its matrix (plasma) is fluid, so blood can flow easily.
- RBCs, WBCs and platelets are formed in bone marrow.
- Blood is essential for transportation, protection and regulation in the body.



REMEMBER: Fluid matrix (plasma) + Specialised cells (RBCs, WBCs, platelets)
= Fluid connective tissue (**Blood**)

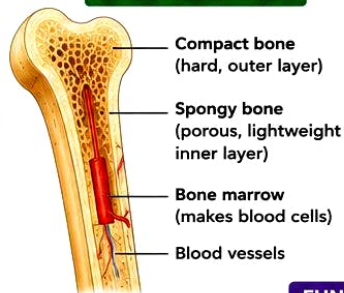
CONNECTIVE TISSUES: BONE AND CARTILAGE

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A. BONE – A RIGID CONNECTIVE TISSUE

Bone is a hard, rigid connective tissue that forms the skeletal framework of the body.

STRUCTURE OF BONE



FEATURES

- Matrix is hard and rigid.
- Made of calcium and phosphorus salts embedded in a network of collagen fibres.
- Contains bone cells (osteocytes) in small spaces (lacunae).
- Bone marrow in certain bones produces blood cells.

FUNCTIONS



Provides shape, support and protection to the body.



Forms a framework with muscles and helps in movement.



Bone marrow produces blood cells.

LOCATIONS & EXAMPLES



Found in: Skull, ribs, vertebrae, limb bones (e.g., femur, humerus) and other parts of the skeleton.

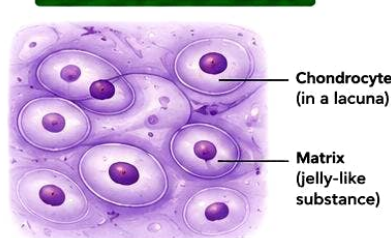
Q. Why are bones hard while blood, from the same tissue category, is a liquid?

A. Bone matrix is solid and mineralised (calcium/phosphorus + collagen) giving hardness and rigidity, whereas blood matrix (plasma) is a fluid.

B. CARTILAGE – A FLEXIBLE CONNECTIVE TISSUE

Cartilage is a flexible connective tissue that provides support and acts as a shock absorber.

STRUCTURE OF CARTILAGE



FEATURES

- Matrix is firm, smooth and jelly-like.
- Made of protein fibres and a gel-like substance.
- Contains cells called chondrocytes in lacunae.
- Not as hard as bone, but flexible.

FUNCTIONS



Provides flexibility and cushioning.



Supports and maintains shape of structures.



Acts as a shock absorber where bones meet.

LOCATIONS & EXAMPLES



Tip of the nose



Outer ear



Between vertebrae (intervertebral discs)



Ends of long bones (joints)



Trachea (windpipe)

Q. Why does pressing your nose or folding your ear not cause pain or breakage, while bending a bone would?

A. Nose and ear are supported by cartilage, which is soft, flexible and returns to its original shape, unlike bone which is rigid.



KEY CONCEPT: Both bone and cartilage are connective tissues, but their matrices are different.
Hard matrix in bone → rigid and strong | Jelly-like matrix in cartilage → flexible and smooth



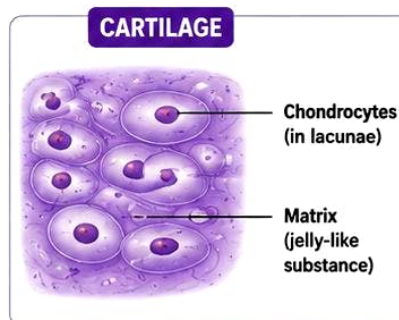
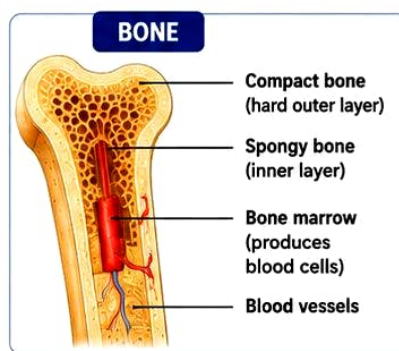
CONNECTIVE TISSUES: BONE vs CARTILAGE

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Bone and cartilage are two types of connective tissues.
They differ in their structure and function.



FEATURE	BONE	CARTILAGE
 Nature	Hard and rigid connective tissue	Soft, flexible connective tissue
 Matrix	Hard, mineralised matrix (calcium & phosphorus salts + collagen fibres)	Firm, jelly-like matrix with protein fibres
 Cells	Osteocytes	Chondrocytes
 Function	Provides support, protection and helps in movement	Provides flexibility, cushioning and absorbs shock
 Blood Supply	Rich blood supply	No direct blood supply (nutrients diffuse from surrounding tissues)
 Location	Skull, ribs, vertebrae, limb bones (e.g., femur, humerus) and other parts of skeleton	Tip of nose, outer ear, trachea (windpipe), ends of long bones (joints), between vertebrae (intervertebral discs)
 Flexibility	Very little (almost rigid)	High (flexible and elastic)



NCERT KEY POINTS

- ✓ Bone forms the rigid framework of the body and protects internal organs.
- ✓ Cartilage is softer than bone and prevents friction between bones at joints.



THINK & ANSWER

Why can you bend your ear but not your arm bone?

Answer: The ear contains cartilage, which is flexible, whereas the arm contains bone, which is hard and rigid.



REMEMBER

Bone



= Strength
& Support

Cartilage

= Flexibility
& Cushioning





CONNECTIVE TISSUES: TENDONS AND LIGAMENTS

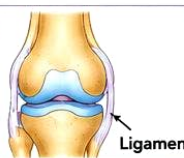
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Tendons and ligaments are **dense fibrous connective tissues**.
They provide strong connection and stability in our body.

TENDONS

 DEFINITION	Tendons are tough, fibrous connective tissues that connect muscles to bones .
 STRUCTURE	- Made up of densely packed collagen fibres arranged in parallel bundles. - Very strong and have great tensile strength. - Little ground substance; no blood vessels. 
 FUNCTION	- Transmit the force of muscle contraction to bones. - Help in movement of body parts.
 EXAMPLES / LOCATION	- Tendon of biceps (connects biceps muscle to radius bone of forearm). - Tendon of quadriceps (connects thigh muscle to tibia through patella). 

LIGAMENTS

 DEFINITION	Ligaments are tough, fibrous connective tissues that connect bone to bone at joints.
 STRUCTURE	- Made up of densely packed collagen fibres arranged in parallel bundles. - Very strong but slightly more elastic than tendons. - Little ground substance; no blood vessels. 
 FUNCTION	- Connect bones to bones at joints. - Hold the bones together and provide stability. - Prevent excessive or abnormal movement at joints.
 EXAMPLES / LOCATION	- Ligaments of knee joint (e.g., anterior cruciate ligament). - Ligaments in the ankle, shoulder, elbow joints. 

DIFFERENCES AT A GLANCE

Feature	Tendons	Ligaments
 Connects	Muscle to bone	Bone to bone
 Function	Transmit force of muscle contraction and bring about movement	Hold bones together and provide stability to joints
 Elasticity	Less elastic	More elastic than tendons
 Location	Outside the joint	Inside the joint



KEY POINTS

- ✓ Both are made of dense collagen fibres.
- ✓ They are strong tissues and resist stretching.
- ✓ They work together to allow smooth and safe movements.



 **REMEMBER:** TENDON connects **muscle to bone** and moves the body. | LIGAMENT connects **bone to bone** and stabilizes the joint.